Python Tsunami

Python I - An Introduction to Python

Who are we?

Center for Health Data Science

Center for Health Data Science (HeaDS)

Section for Health Data Science and Artificial Intelligence (HDSAI)

The mission of HeaDS is to strengthen health data science research at SUND:

- Visible hub for Health Data Science
- Research within the field of HDS
- SUND Data Lab

https://heads.ku.dk

SUND Data Lab

COURSES & WORKSHOPS

Data Science skills, Tools and HDS Topics

COMMISSIONED RESEARCH

Commissioned Data Science Analysis
Commissioned Supervision

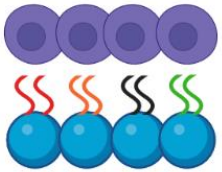
CONSULTATIONS

Need guidance? Drop by for a consultation on your research project

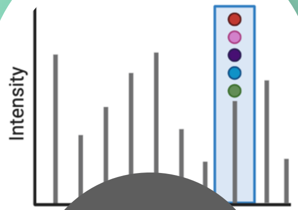
MATCHMAKING

Conference, seminars and networking events - Join us!

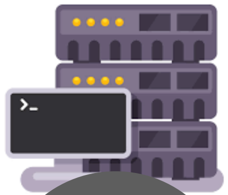
Commissioned Research



ANALYSIS OF SINGLE CELL RNA SEQ



ANALYSIS OF BULK OMICS DATA



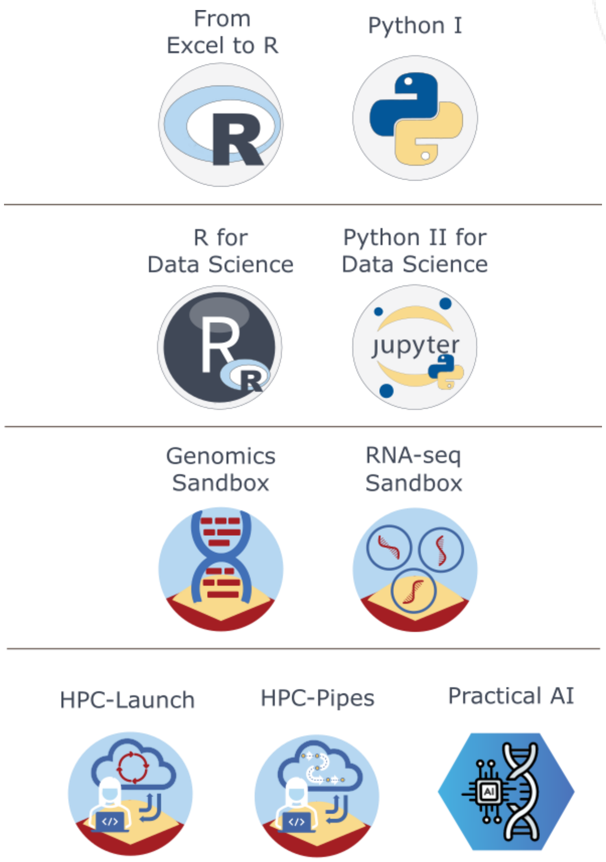
BIG DATA MANAGEMENT - HPC NEEDED

- Machine Learning
- Biometric Data
- Metagenomics
- Epigenetics
- Biological Networks
- Seq. Motif Discovery
- Text mining
- *Biostatistical Models*

HeaDS Courses

Skill Sets at Different Levels:

- Programming & Data Science (R, Python)
- Biomedical Data Analysis
- Advanced Computational Workflows



The Python Tsunami Team

The Team:

Thilde Terkelsen

Suze Roostee

Iñigo Prada

Annike Kjølby Kristensen

Astrid Filt

Original Content:
Henrike Zschach (Research IT)
Alberto Delgardo (DTU, Biosustain),
Henry Webel (DTU, Biosustain),
Annelaura Nielsen (University Hospital),
Rita Colaço (PRI)

About this course

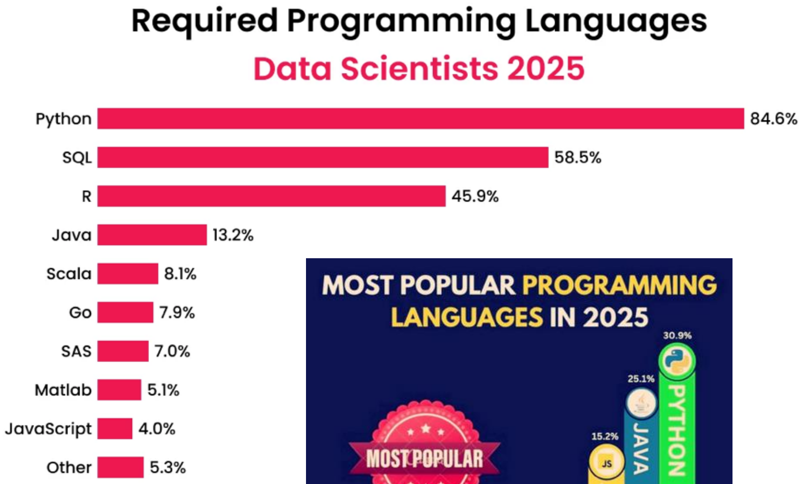
Day 1		Day 2		Day 3	
08:45-09:00	Morning Coffee	08:45-9:00	Morning Coffee	08:45-09:00	Morning Coffee
09:00-09:30	Intro to Python & Colab	09:00-10:00	Basics recap quiz + Q&A	09:00-09:30	Pandas recap quiz + Q&A
09:30-10:10	Variables & Data Types	10:00-10:30	Pandas: Series, Dataframes, Renaming	09:30-10:30	Visualization
10:10-10:20	Break	10:30-10:45	Break		
10:20-11:00	Iterables I: Lists	10:45-11:30	Pandas: Indexing, Selection & Summary Functions	10:30-10:45	Break
11:00-11:10	Break			10:45-12:00	Visualization
11:10-12:00	Iterables II: Sets, Dicts & Tuples	11:30-12:00	Pandas: Modifying Data		
12:00-13:00	Lunch	12:00-13:00	Lunch	12:00-13:00	Lunch
13:00-14:15	Booleans, Operators & Conditions	13:00-14:15	Pandas: Groupby, Sort & Impute	13:00-13:30	Local Installations
14:15-14:30	Break	14:15-14:30	Break	13:30-16:00	Dataset Exercise
14:30-16:00	Loops	14:30-15:15	Pandas: Handling missing data + Pandas Q&A		
16:00	See you tomorrow!	15:15	See you tomorrow!	16:00	Course Finished!

Python

Why Python?

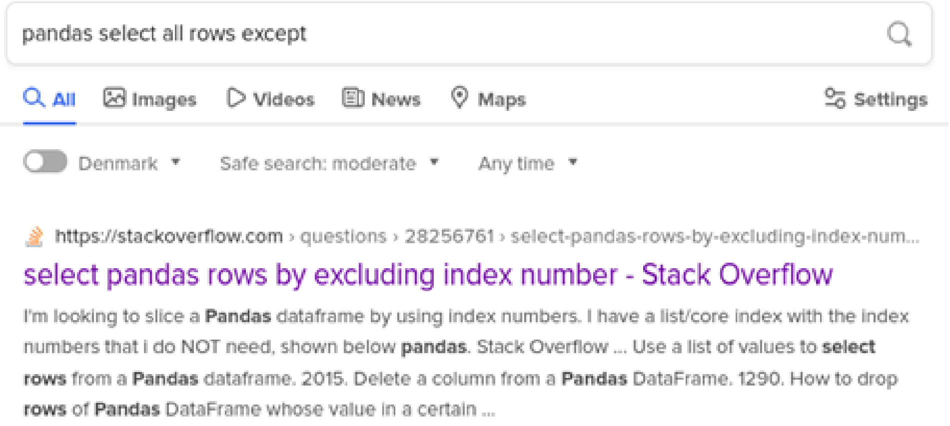
Python is a great programming language for both beginners and advanced programmers:

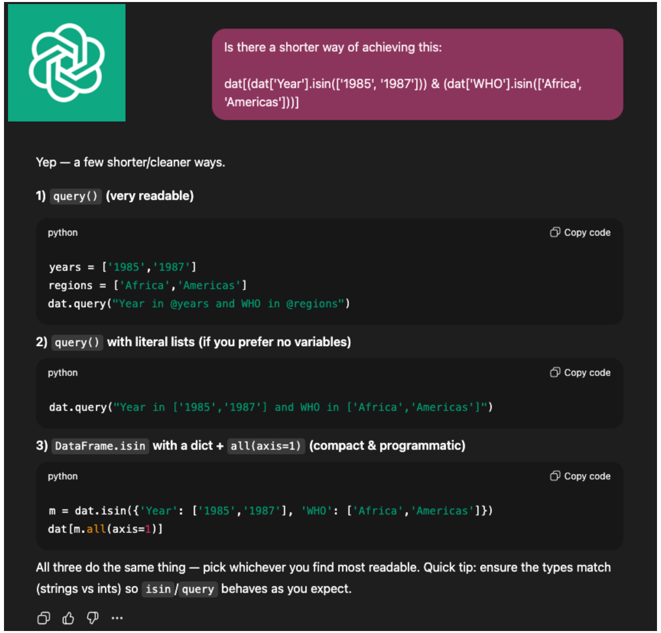
- Easy to grasp, close to natural language
- Many learning resources available
- Large community
- Libraries / Packages
- Makes machine learning 'easy'



Where to get help

Online communities such as **stackoverflow** are important tools in programming.





Or ask **an AI tool.** But remember: “With great power comes great responsibility!”

Python Environments

Google Colab

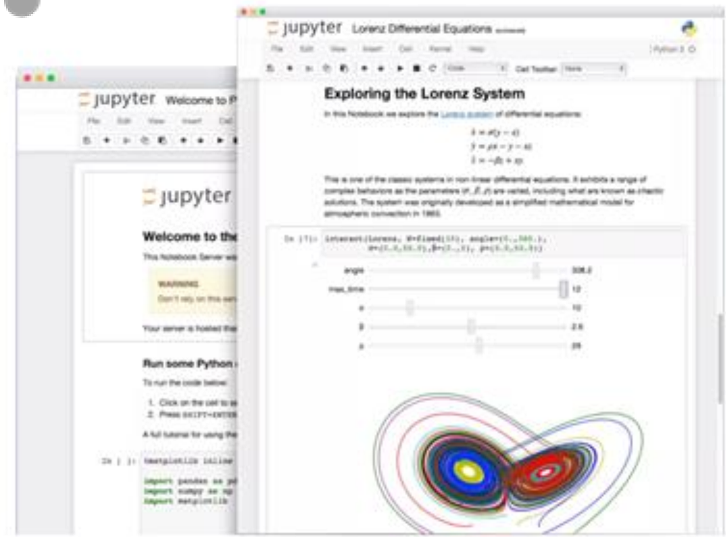
Google Colab is a Jupyter Notebook hosted on Google's servers, not your own machine. It still runs in your browser.

- Tool to write, execute and share Python code through the browser
- Requires no setup to use and provides free access to computing resources on Google's servers including GPUs
- Connected to a Google account and data and notebooks can be accessed through Google Drive.

We'll use Colab during the course.

Jupyter notebook

Jupyter Notebook is an **open-source application** to create and share documents that contain code, visualizations and text (markdown).



- Browser-based development environment for creating, running and sharing Python code
- Combine code with text and output – generates; html, pdf, website
- Runs from your **local installation**. I.e. you need Python and the libraries you want to use installed on your computer

Jupyter notebook

Text cell

```
> A variable is simply a name that holds a value.

> Variable **containers** --> name and value

* Create variables by assigning a value to a name (just like using variables in
math).
* Variable names should be meaningful, i.e. not just ``a``, ``b``, ``c``
* Variables are always **assigned** with the variable name on the left and the
value on the right of the ***equals*** sign. For instance:

    * `a_variable = 100`
    * assigned to other variables: `another_variable = a_variable`
    * reassigned at any time: `a_variable = 435`
    * assigning several variables at the same time: `all, at, once = 1, 130, 43`

* Variables **must** be assigned before they can be used.
```

> A variable is simply a name that holds a value.

> Variable **containers** --> name and value

- Create variables by assigning a value to a name (just like using variables in math).
- Variable names should be meaningful, i.e. not just `a`, `b`, `c`
- Variables are always **assigned** with the variable name on the left and the value on the right of the ***equals*** sign. For instance:
    - `a_variable = 100`
    - assigned to other variables: `another_variable = a_variable`
    - reassigned at any time: `a_variable = 435`
    - assigning several variables at the same time: `all, at, once = 1, 130, 43`
- Variables **must** be assigned before they can be used.

```python
# Create a variable x by assigning a value to x
x = 100
print(x)

# What is the container and what are the data here?
```

Code cell

Jupyter notebook

Getting help inside the notebook

```
[5] help(print)

Help on built-in function print in module builtins:

print(*args, sep=' ', end='\n', file=None, flush=False)
    Prints the values to a stream, or to sys.stdout by default.

    sep
      string inserted between values, default a space.
    end
      string appended after the last value, default a newline.
    file
      a file-like object (stream); defaults to the current sys.stdout.
    flush
      whether to forcibly flush the stream.
```

If you don't like AI to do the work for you

(Colab specific)

Settings

- Site
- Editor
- **AI assistance**
- Colab Pro
- GitHub
- Miscellaneous

- [x] Show AI-powered inline completions
- [ ] Consented to use generative AI features
- [ ] Hide generative AI features

Gemini can make mistakes, so double-check responses and use code with caution.

Python Libraries

Libraries

Python has many libraries, also called packages, that other programmers have developed. Find and **use** them!

Well-maintained libraries generally are:

- Tested
- Optimized
- Documented

During this course we will use:

- Pandas (all the data analysis)
- Math (basic math)
- Seaborn (visualization)

Libraries

Running Python from a local installation requires you to have libraries **downloaded & installed** before you can use them.

On **Google Colab** you can generally just import, they are already installed.

- Import the math library:

```
import math
```

- Now I can use functions from that library, i.e. calculating the logarithm or square root:

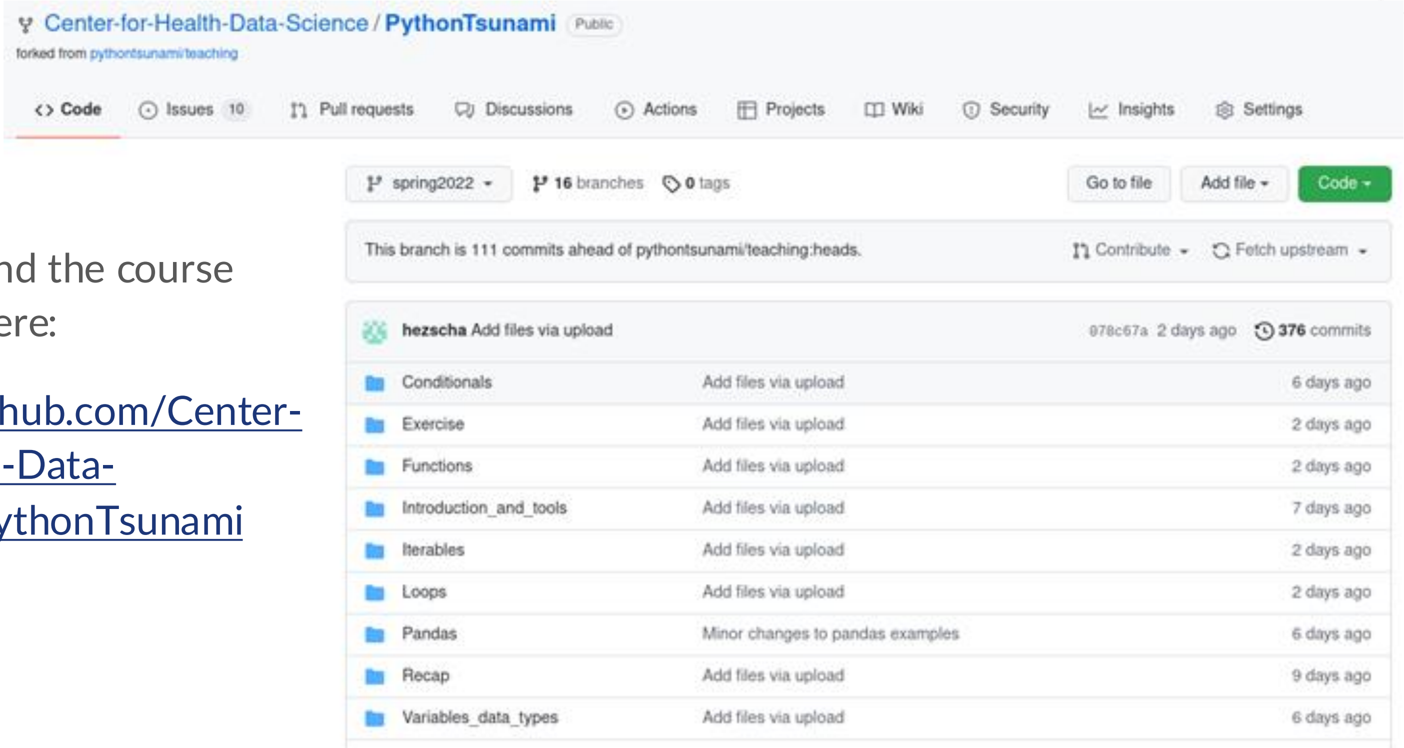
```
math.log(3)

math.sqrt(4)
```

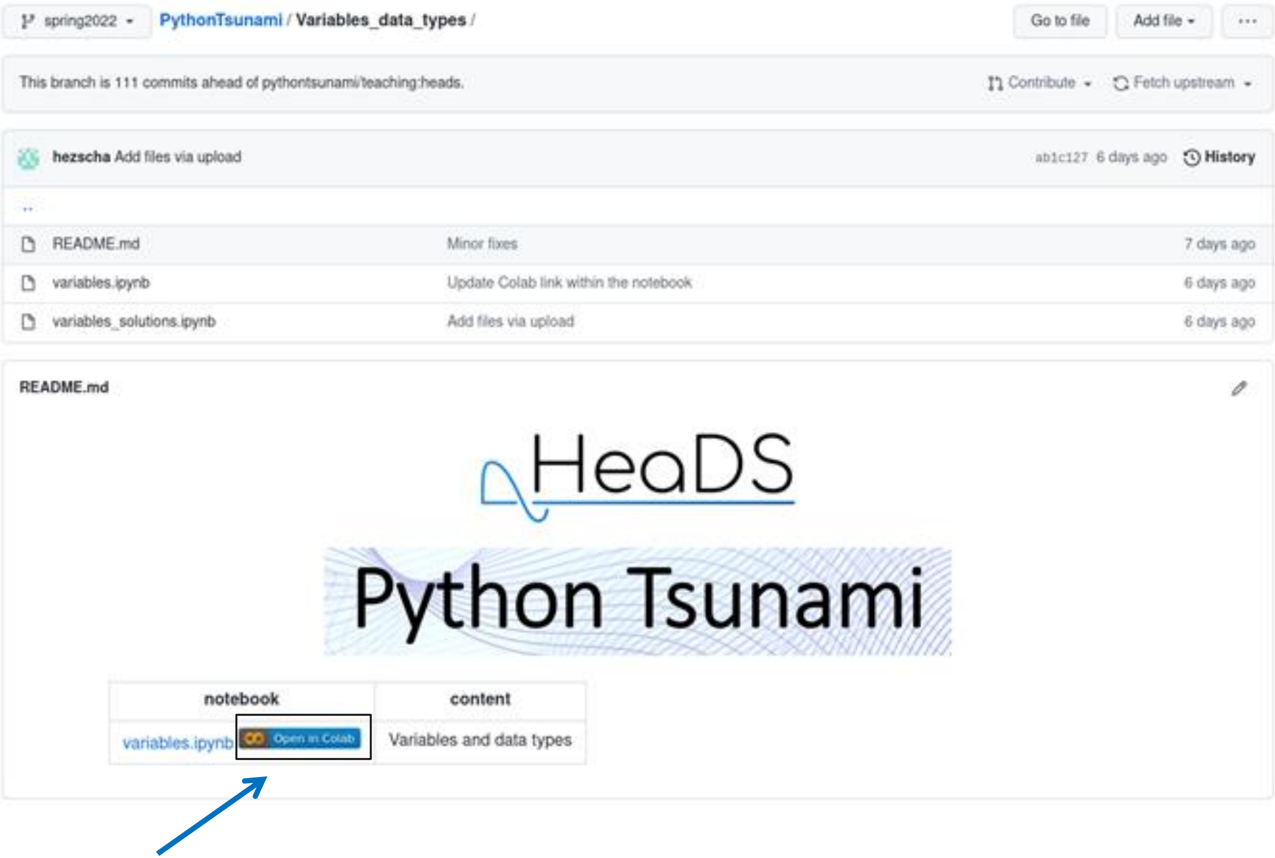
Course Material

You can find the course material here:

https://github.com/Center-for-Health-Data-Science/PythonTsunami

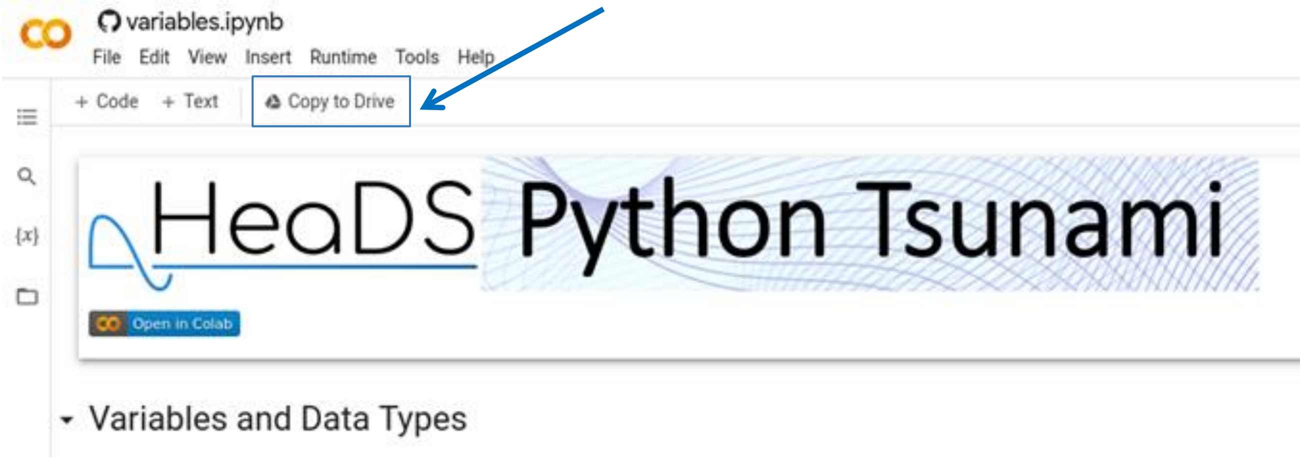


Course Material



Course Material

Remember to **save a copy** to your own google drive so you can save your notes and exercises!



Short Introduction

Take the next 5 mins to introduce yourself at your table:

- Name
- Position/Unit
- What you do (very briefly!)
- Why would you like to learn Python?